

HRIDAYESH KUNDU

Course : **M.Sc. (Hons.)**, Chemistry and **B.E. (Hons.)**, Electrical and Electronics Engineering, 2028

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CGPA : 8.96

ACADEMIC DETAILS

COURSE	INSTITUTE/COLLEGE	BOARD/UNIVERSITY	SCORE	YEAR
CLASS XII	Vivekananda Academy, Serampore	CBSE	87.2 %	2022
CLASS X	Gospel Home School	ICSE	95 %	2020

Subjects / Electives	Digital Design, Signals and Systems, MicroElectronics, MicroProcessors and Interfacing, Computer Architecture, Reconfigurable Computing, Electronic Devices
Technical Proficiency	ModelSim, TL-Verilog, System Verilog, Zynq-7000, MATLAB, Verilog, Cadence Virtuoso, Fusion 360, Vivado Design Suite

INTERNSHIPS

Hardware Development Intern, Genesis Gas Solutions Pvt. Ltd. **May 2025 - Jul 2025**

- Developed an IoT modem prototype using LoRa communication for wireless data transmission.
- Integrated an Arduino Uno with multiple sensors to implement and validate the communication system.

PROJECTS

APB Slave Verification Environment - System Verilog (Ongoing) - Verification **Oct 2026 - Present**

- Developing a class-based layered testbench in **SystemVerilog** for an APB slave, structured as transaction, generator, driver, monitor and scoreboard components connected through mailboxes.
- Targeting **constrained-random** read/write stimulus checked against a reference memory model, with functional coverage and assertions on the APB handshake as next steps.

Fixed-Point Quadratic Equation Root Solver - FPGA / RTL Design (Zynq-7000) **Sep 2026 - Oct 2026**

- Designed a hardware solver in Verilog computing real and complex roots of a quadratic equation, with inputs and roots in signed **Q16.16 fixed-point.**
- Implemented iterative **Newton-Raphson** engines for reciprocal (division) and square root, handling fixed-point truncation and normalization without any vendor arithmetic IP.
- Verified functionality on hardware using a **VIO**-based debug flow to drive coefficients and observe results.

Parameterizable UART Transmitter & Receiver - RTL Design **May 2026 - Jul 2026**

- Designed and implemented parameterizable UART Transmitter and Receiver modules in Verilog.
- Developed **FSM**-based serial communication logic with configurable parity and stop-bit support.
- Verified UART functionality using simulation testbenches across multiple transmission scenarios.

Biocoin Glucose Monitoring Analysis - Hardware Research **Sep 2026 - Present**

- Analyzing the Biocoin open-source biomedical device platform to identify hardware limitations in its glucose monitoring device.
- Studying the sensor front-end and signal acquisition chain to propose improvements in accuracy and power.

EXTRA CURRICULAR ACTIVITIES

Chess

Represented **BITS Goa** in inter-campus chess tournaments, including **Arena** (BITS Hyderabad) and **Spree** (BITS Goa).

SCHOLARSHIPS

Merit Cum Need Scholarship **Jul 2026**

I am recipient of 80% scholarship based on my Merit in academics and Annual Family Income.

Samsung Fellowship **May 2026**

Awarded Samsung Fellowship in Grade II (75%) for ISWDP programme conducted by Synopsys, IISc Bangalore and Samsung.

LANGUAGES KNOWN

Hindi, Bengali, English